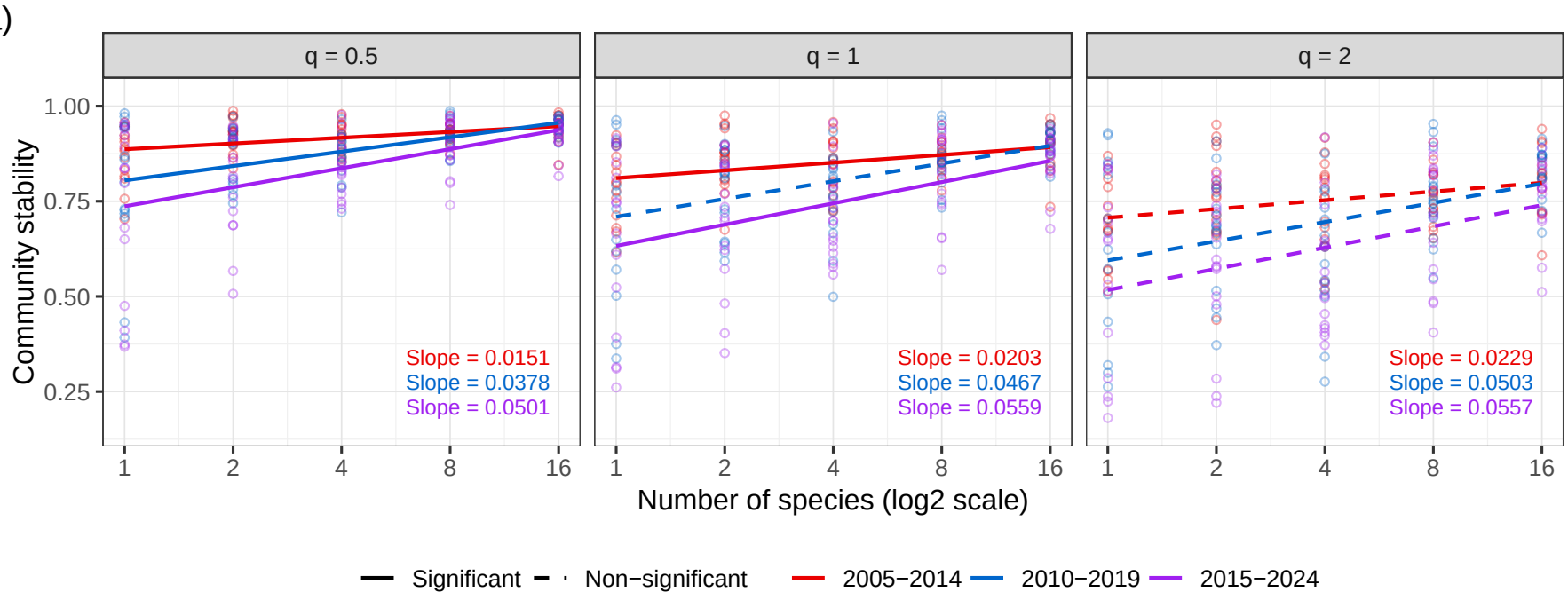
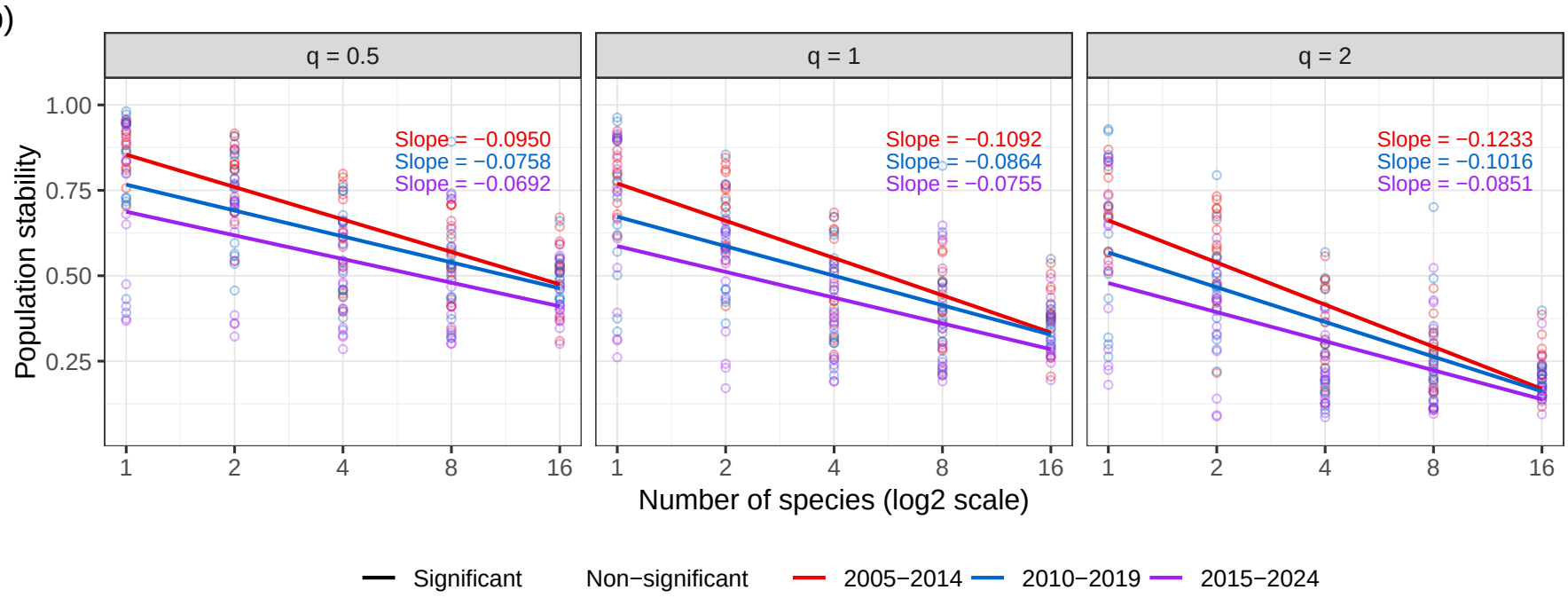
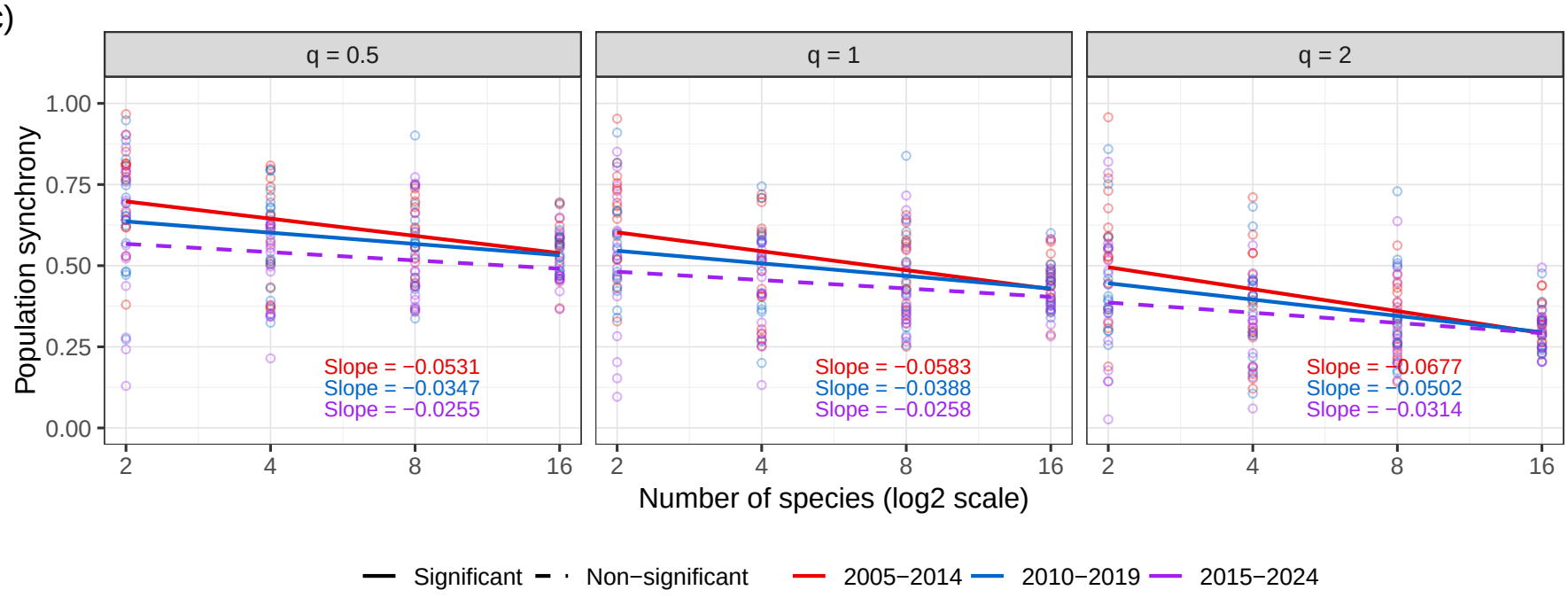
(a)


q = 0.5
q = 1
q = 2
Community stability
1.00
0.75
0.50
0.25
Slope = 0.0151
Slope = 0.0378
Slope = 0.0501
Slope = 0.0203
Slope = 0.0467
Slope = 0.0559
Slope = 0.0229
Slope = 0.0503
Slope = 0.0557
1
2
4
8
16
Number of species (log2 scale)
Significant
Non−significant
2005−2014
2010−2019
2015−2024

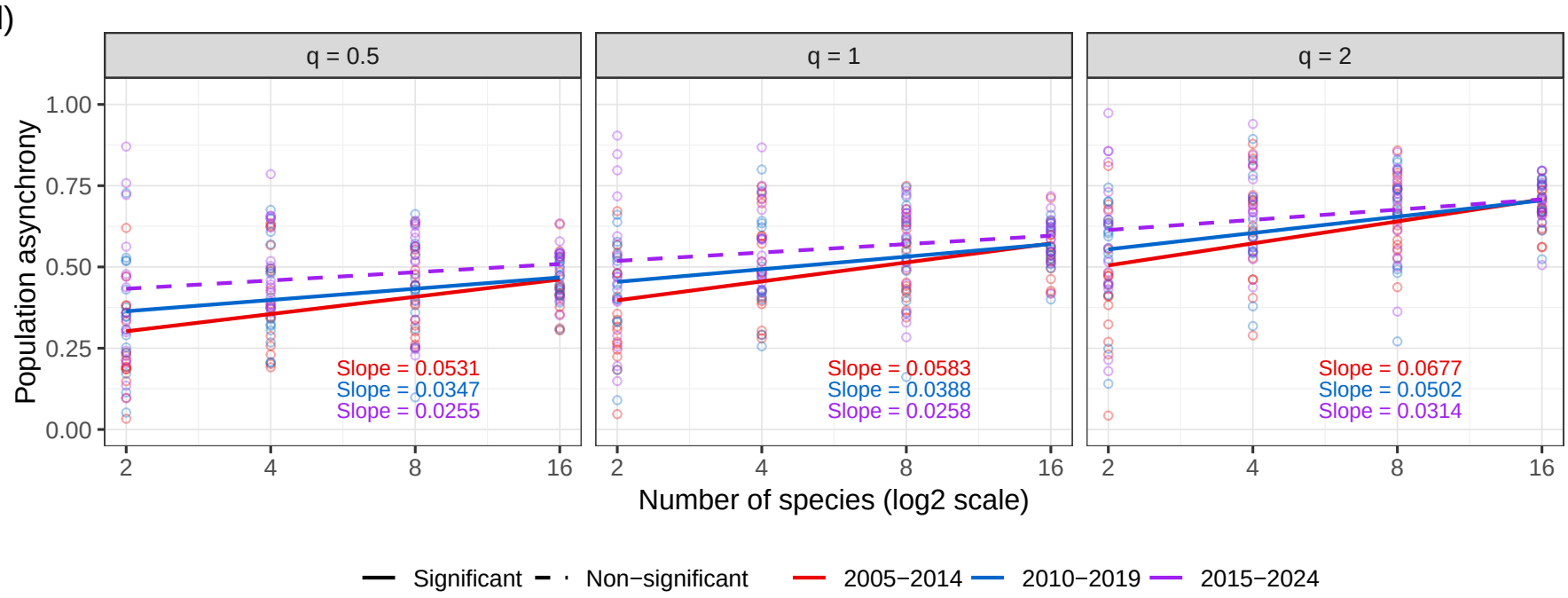
(b)


q = 0.5
q = 1
q = 2
Population stability
1.00
0.75
0.50
0.25
Slope = −0.0950
Slope = −0.0758
Slope = −0.0692
Slope = −0.1092
Slope = −0.0864
Slope = −0.0755
Slope = −0.1233
Slope = −0.1016
Slope = −0.0851
1
2
4
8
16
Number of species (log2 scale)
Significant
Non−significant
2005−2014
2010−2019
2015−2024

(c)


q = 0.5
q = 1
q = 2
Population synchrony
1.00
0.75
0.50
0.25
0.00
Slope = −0.0531
Slope = −0.0347
Slope = −0.0255
Slope = −0.0583
Slope = −0.0388
Slope = −0.0258
Slope = −0.0677
Slope = −0.0502
Slope = −0.0314
2
4
8
16
Number of species (log2 scale)
Significant
Non−significant
2005−2014
2010−2019
2015−2024

(d)


q = 0.5
q = 1
q = 2
Population asynchrony
1.00
0.75
0.50
0.25
0.00
Slope = 0.0531
Slope = 0.0347
Slope = 0.0255
Slope = 0.0583
Slope = 0.0388
Slope = 0.0258
Slope = 0.0677
Slope = 0.0502
Slope = 0.0314
2
4
8
16
Number of species (log2 scale)
Significant
Non−significant
2005−2014
2010−2019
2015−2024